6 b output.sci (C:\Users\LENOVO\Documents\6 b output.sci) - SciNotes

*6 b output.sci

```
// Test Case: Analysis of a Discrete-Time System
// Define the input signal x[n]
n = 0:20; // Discrete time index
x_n = (n >= 0) - (n >= 10); // Example input: A rectangular pulse from n=0 to n=9
// Initialize the output signal y[n] with zeros
y_n = zeros(1, length(n));
// Define the system coefficients
b = [0.5, 0.3, 0.2]; // Coefficients of x[n], x[n-1], and x[n-2]
// Compute the output y[n] using the difference equation
for i = 3:length(n)
y_n(i) = b(1) * x_n(i) + b(2) * x_n(i-1) + b(3) * x_n(i-2);
end

// Plot the input signal x[n]
clf; // Clear the current figure
subplot(2,1,1);
bar(n, x_n, 'b');
xlabel('n (Discrete Time Index)');
ylabel('x[n]');
title('Input Signal x[n]');
// Plot the output signal y[n]
subplot(2,1,2);
bar(n, y_n, 'r');
xlabel('n (Discrete Time Index)');
ylabel('y[n]');
title('Output Signal y[n] of the System');
```

Graphic window number 0

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